

Kia/Hyundai Theft: Assignment Summary

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Project Overview

This data visualization project analyzed Kia and Hyundai vehicle theft data to examine whether recent increases represent a distinct, targeted public safety issue rather than a general rise in auto theft. The analysis was designed as a municipal public safety briefing to support evidence-based decision-making by city officials.

Audience, Purpose, and Medium

Audience:

City officials and public safety leadership who possess familiarity with crime statistics, trend reporting, and policy-driven decision-making. This professional audience requires clear, actionable insights that support resource allocation, prevention planning, and program approval rather than exploratory or technical analysis.

Purpose:

To evaluate the scope, growth, and geographic concentration of Kia and Hyundai vehicle thefts and determine whether targeted intervention strategies are warranted. The call to action requests approval for focused prevention measures, including steering-wheel lock distribution, adjusted patrol allocation, and localized public awareness campaigns in high-impact jurisdictions.

Medium:

A professional slide presentation was selected as the primary medium to align with

executive briefing and policy review norms. This format supports sequential narrative development, integrates multiple visualization types, and facilitates structured discussion and decision-making. All visualizations were created in R and exported for presentation use.

Design Strategy

Color Palette:

A restrained, high-contrast palette was used to consistently distinguish Kia and Hyundai thefts from other vehicle thefts. Neutral grays provided contextual baselines, while red tones highlighted areas of elevated risk and growth without introducing unnecessary visual emphasis.

Typography:

Clean, legible fonts were used with clear hierarchy to prioritize key trends and comparisons. Emphasis was placed on readability and efficient information consumption in both live briefings and asynchronous review settings.

Layout:

A data-forward layout positioned visualizations as the primary communication elements, supported by concise explanatory text. This approach aligns with executive expectations and minimizes narrative overhead while maintaining interpretive clarity.

Spacing and Alignment:

Consistent spacing and alignment were applied across all visuals to reduce cognitive load and support rapid comparison between charts. Gestalt principles, including similarity and proximity, were used to reinforce categorical distinctions and temporal patterns.

Ethical Considerations

Data Transformations:

Datasets required structural cleaning, including reshaping wide-format data into tidy structures and converting numeric values stored as text into usable numeric formats. Aggregations were performed at the city and time-period level without modifying individual records.

Legal and Regulatory Compliance:

All datasets were obtained from publicly available sources, including municipal records and journalism-based reporting compiled by Motherboard and VICE News. Use of the data complies with public data access and reuse guidelines.

Presentation Risks:

Visualizations were designed to avoid overstating causality or generalizing findings beyond the available data. Milwaukee was explicitly labeled as a case study to prevent misinterpretation as a representative city.

Filtering and Assumptions:

Filtering was applied to exclude records with missing or invalid values and to focus selected visuals on top-impact cities for readability. These choices were made transparently and did not alter underlying trends.

Data Credibility and Ethical Acquisition:

Data sources were cross-referenced across multiple datasets to validate consistency in reported patterns. Differences in reporting practices across jurisdictions were acknowledged as a limitation. All data were sourced through publicly accessible records and reporting initiatives, ensuring ethical acquisition without proprietary access or confidential information.

Risk Mitigation:

Potential ethical implications were addressed through transparent documentation of transformations, conservative framing of findings, clear labeling of case studies, and acknowledgment of data limitations that may influence interpretation.